

Global Renewable Energy in 2025

How Far Have We Come?



A data-driven overview of renewable energy's growing share
in the global energy and electricity mix

Source: Our World in Data — Renewable Energy

Data: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

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May 2025

75% of Global GHG Emissions Still Come from Fossil Fuels



75%

of global greenhouse gas
emissions come from fossil fuels

Burned for energy across all sectors
since the Industrial Revolution



5M+

premature deaths per year
from fossil fuel air pollution

Local air pollution is a major
health crisis alongside climate change



The world must
rapidly shift to
low-carbon sources

Nuclear and renewable technologies
are essential for decarbonization

The Renewable Energy Opportunity

Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources.

Traditional biomass (important in lower-income settings) is typically excluded from modern renewable statistics.

Electricity is the sector where renewables have made the most progress — but transport and heating lag behind.

KEY INSIGHT: Three-quarters of the climate problem is an energy problem — shifting to renewables is not optional, it is essential.

Renewables Now Supply ~14% of Global Primary Energy

~14%

of global primary energy from
renewables (2024, substitution method)



Primary energy includes:

Electricity, transport, and heating combined



Why lower than electricity share:

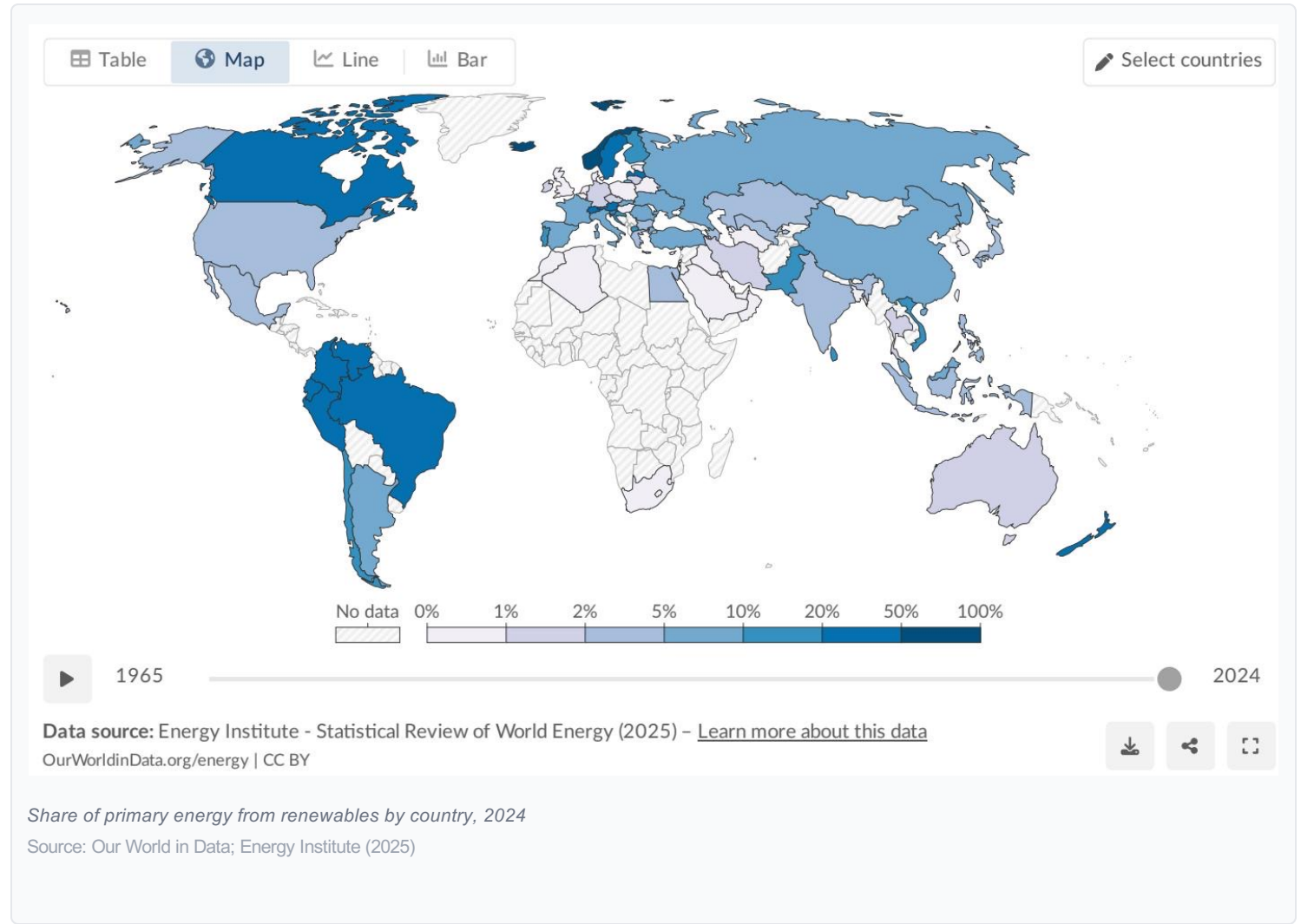
Transport and heating remain harder to
decarbonize — more reliant on oil and gas



Regional variation is significant:

Norway, Brazil, New Zealand: large shares
Middle East, parts of Asia: below 5%

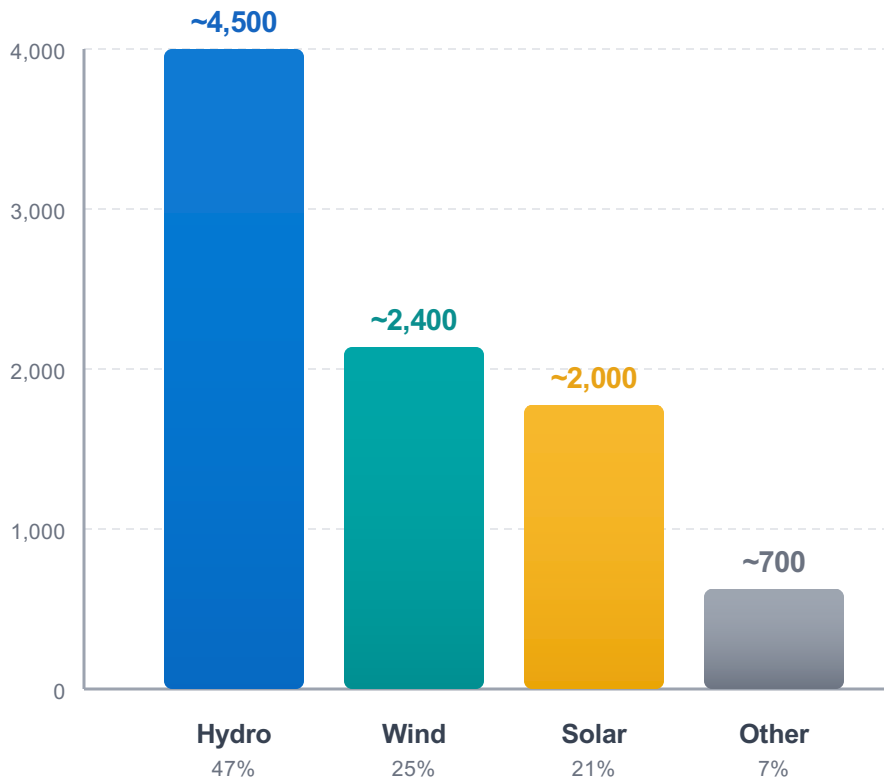
~1/7 of primary energy is renewable



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global Renewable Electricity Generation (2024)

Approximate generation in TWh



TWh = terawatt-hours. Other includes geothermal, biomass, wave, tidal.

Wind + Solar Nearly Match Hydro

Wind (~2,400 TWh) + Solar (~2,000 TWh) = ~4,400 TWh
vs. Hydropower at ~4,500 TWh — closing the gap rapidly

Combined wind+solar share: 46% of renewable electricity

Acceleration Since ~2010

Wind and solar generation has grown dramatically since 2010, driven by falling costs and strong policy support worldwide.

Hydropower Still Leads — But Growth Is Limited

Hydropower generates ~47% of renewable electricity and has operated at scale for over a century. However, its growth potential is limited compared to the rapid expansion of wind and solar.

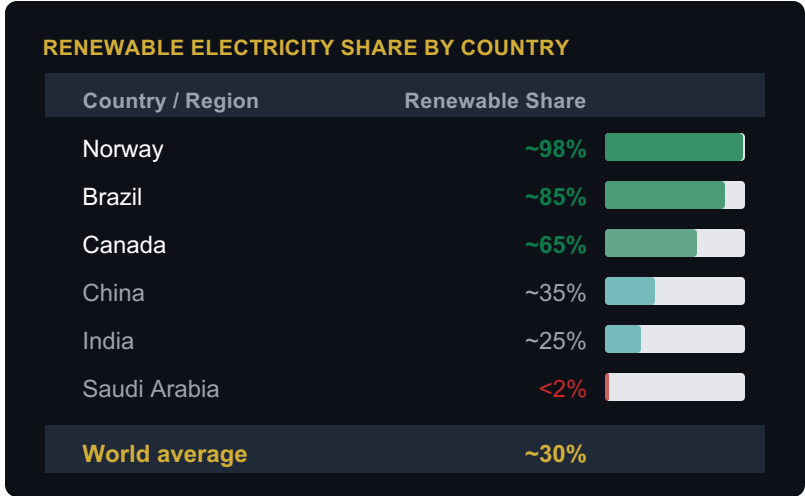
DATA TABLE

Technology	Generation (TWh)	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other renewables	~700	~7%

Almost One-Third of Global Electricity Now Comes from Renewables

~30%

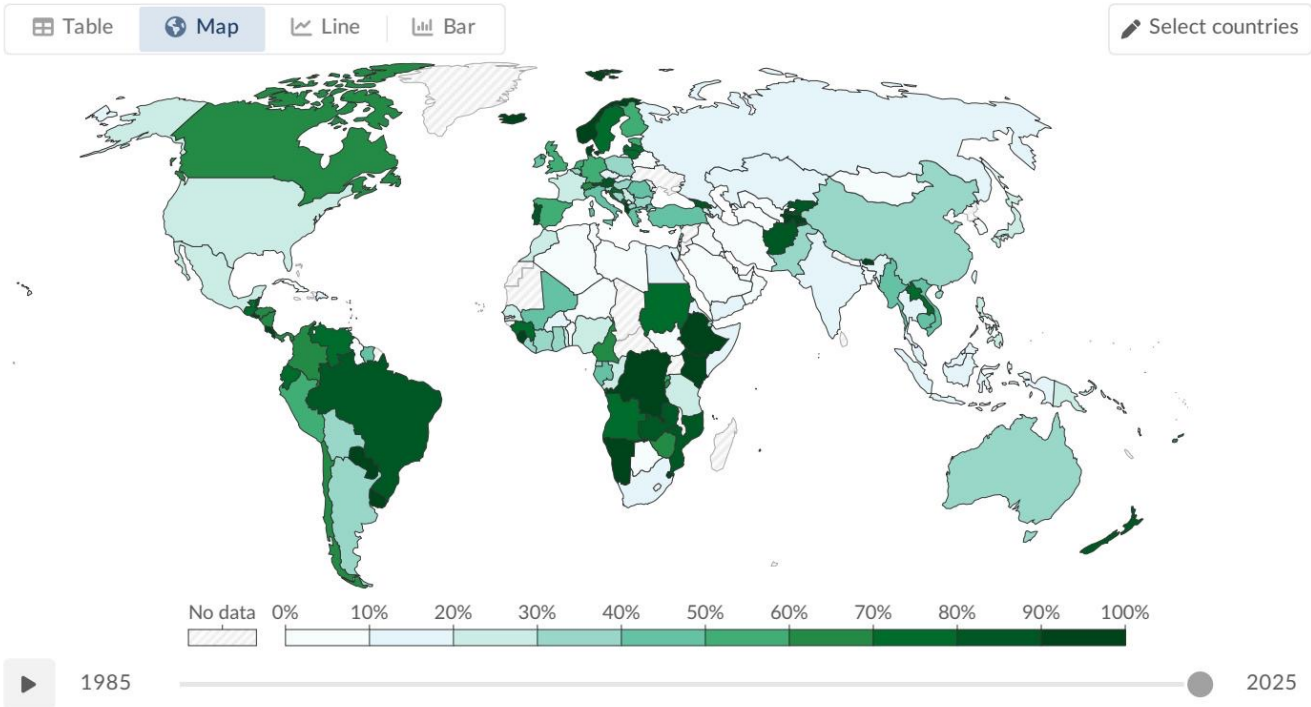
of global electricity from renewable sources (2025)



Several African and Middle Eastern countries remain below 10%.

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Share of electricity from renewables by country, 2025
Source: Our World in Data; Ember (2026)

Hydropower Remains the Largest Renewable Source at ~50% of Generation

~50%

of all renewable electricity
comes from hydropower

✓ Generating at Scale for Over a Century

Hydropower is the most mature renewable technology, with established infrastructure worldwide

▮ Top Producers

China, Brazil, and Canada lead global hydropower generation by a significant margin

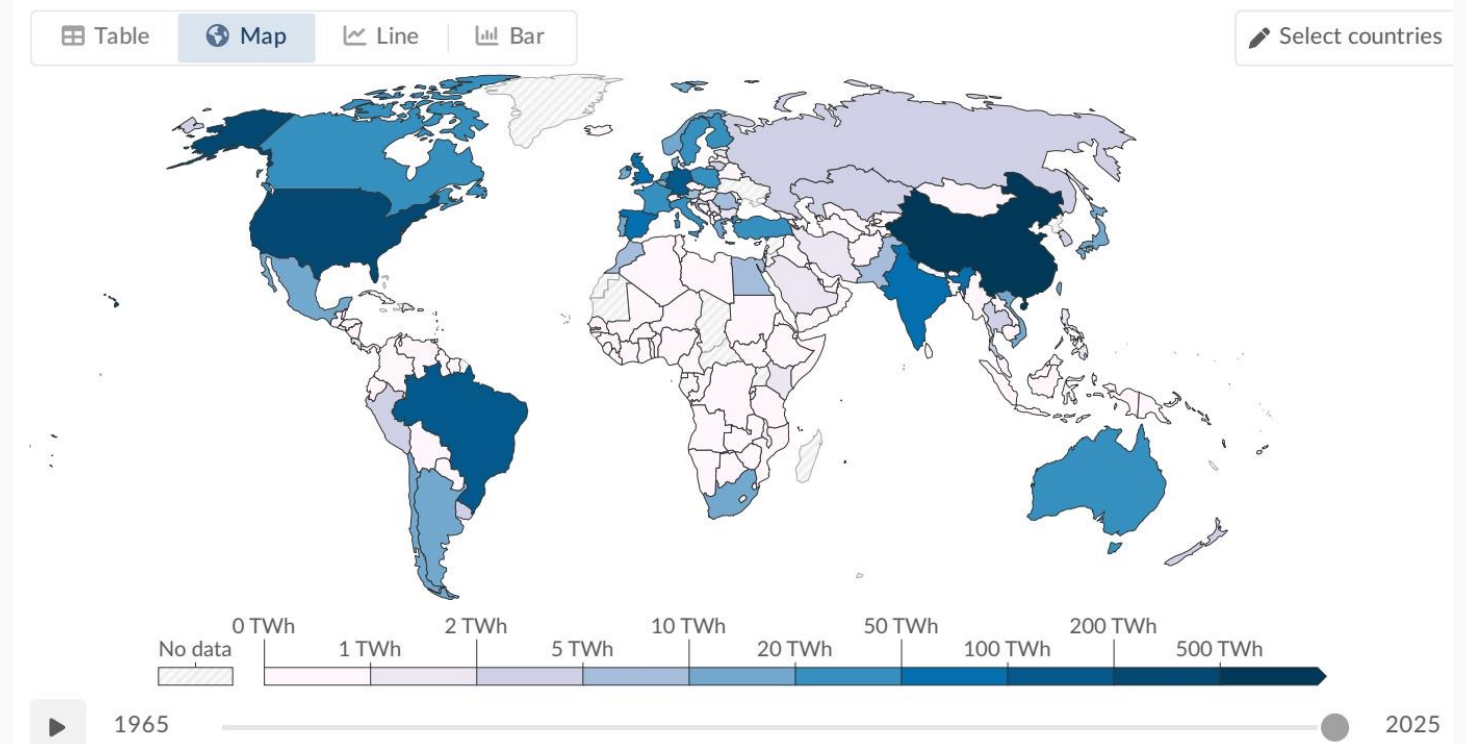
⚠ Growth Potential Is Limited

Most suitable river sites already developed; expansion constrained by geography, ecology, and climate variability (droughts reduce output)

COMPARISON

Hydropower ~4,500 TWh

Wind + Solar combined ~4,400 TWh



Hydropower generation by country, 2025

Source: Our World in Data; Ember (2026)

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind vs. Solar: Generation & Share Comparison

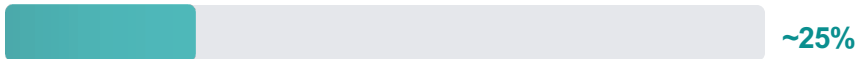
Based on 2024 generation data (not installed capacity)

Wind Energy

Generation (TWh)



Share of renewable electricity



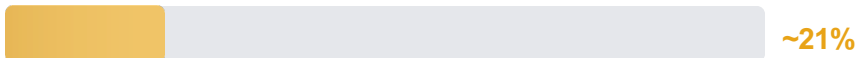
Major contributor in Europe, China, and the US
Includes both onshore and offshore installations

Solar Energy

Generation (TWh)

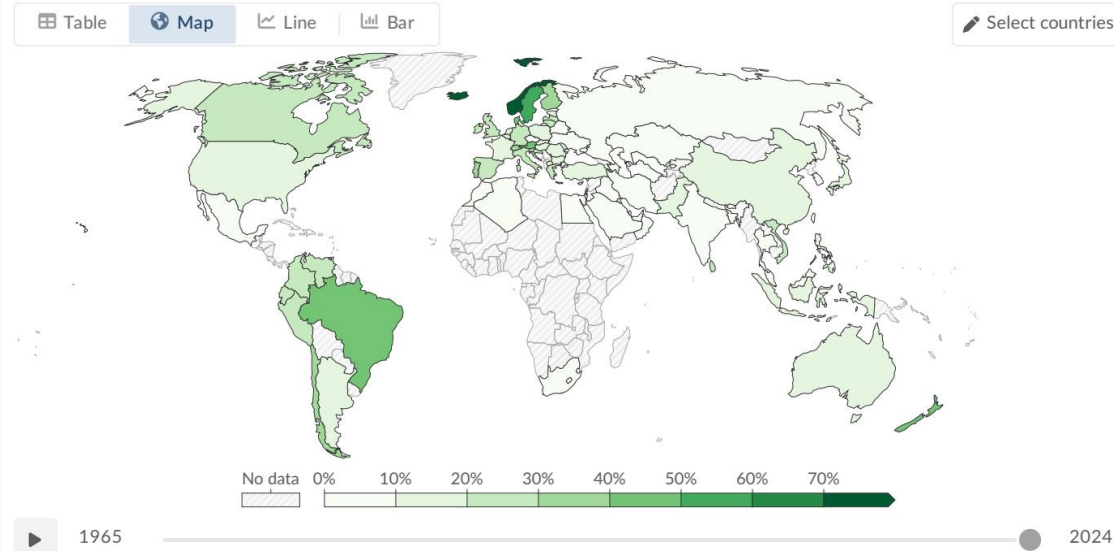


Share of renewable electricity



Fastest-growing energy source globally
Dramatic growth since ~2010; driven by falling costs

COMBINED: ~4,400 TWh — nearly matching hydro's ~4,500 TWh



Wind power generation by country, 2025

Source: Our World in Data; Ember (2026)

Solar is the fastest-growing source in history

Global solar generation has increased dramatically since 2010, driven by falling costs and policy support across major economies.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

LEADERS: 65–98% Renewable Electricity

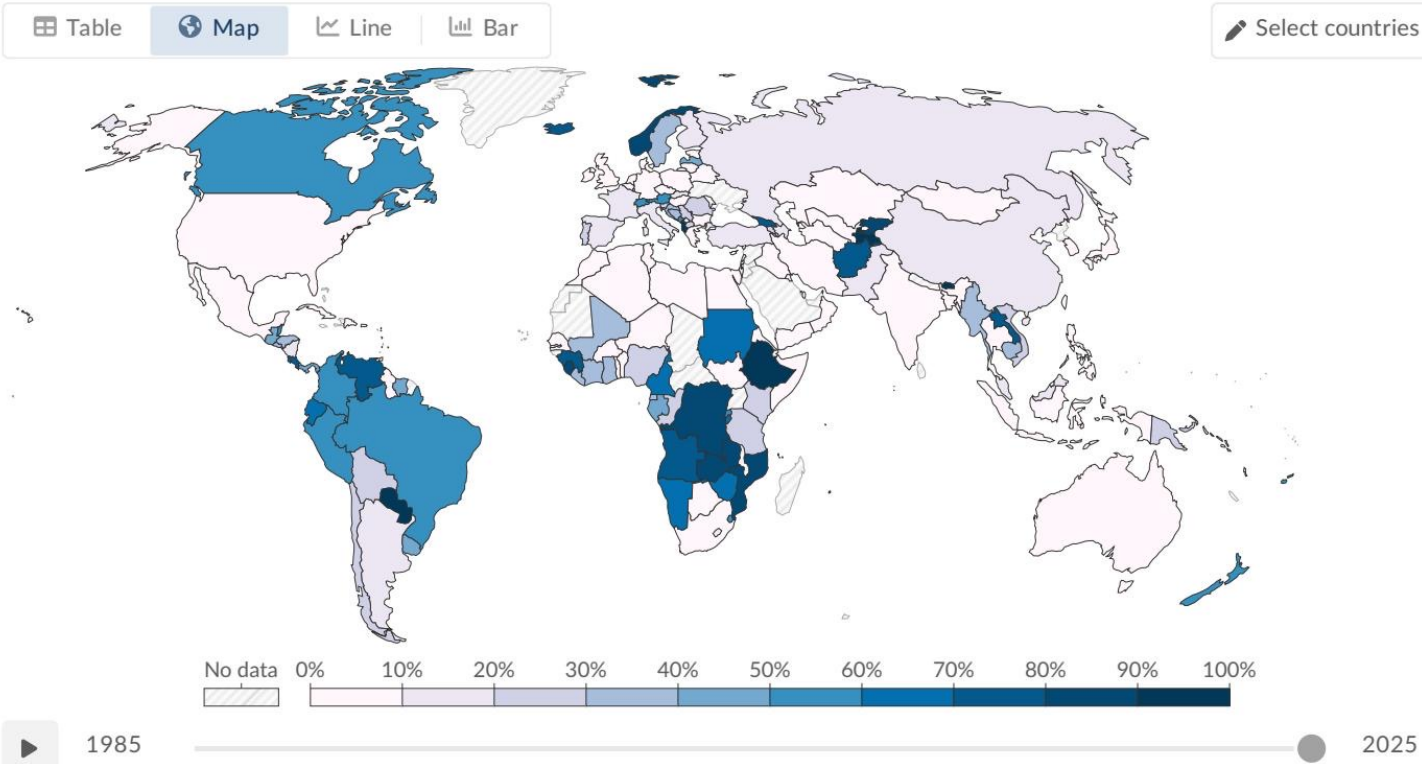
- Scandinavia (65–98%)**
Abundant hydropower drives near-total decarbonization
- Brazil (~85%)**
Large hydroelectric infrastructure (Itaipu, etc.)
- Canada (~65%)**
Vast hydro resources; growing wind/solar investment

INVESTORS: Wind & Solar Pioneers

Denmark and Germany have invested heavily in wind and solar, compensating for limited hydro resources

LAGGARDS: Below 10% Renewable

- Sub-Saharan Africa**
Minimal renewable electricity infrastructure; energy access challenges compound transition barriers
- Middle East**



Share of electricity from hydropower by country, 2025
Source: Our World in Data; Ember (2026)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14% of primary energy and ~30% of electricity now renewable

SIGNIFICANT MILESTONE BUT FAR FROM SUFFICIENT FOR CLIMATE TARGETS



Hydropower remains dominant — generating roughly half of all renewable electricity

But growth potential is limited compared to wind and solar



Wind and solar: fastest-growing energy technologies in history

Combined, they now produce nearly as much electricity as hydropower



Stark regional disparities persist — Scandinavia/Latin America 60–90%+

Middle East and Sub-Saharan Africa remain below 10% renewable electricity



Transport and heating lag far behind electricity

Keeping the overall primary energy share much lower than the electricity share

The pace must accelerate to meet climate targets — every year of delay widens the gap.

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